

Sample Paper 1 — Answer Key

Total: 40 marks (30×1 mark + 5×2 marks), 60 minutes.

Section 1 — Logical Reasoning

1. **D) Sphere** — the other three are flat (2-D) shapes; a sphere is 3-D.
2. **B) 5** — a pentagon has 5 sides.
3. **A) EPH** — the rule is "+1 letter": $D \rightarrow E$, $O \rightarrow P$, $G \rightarrow H$.
4. **C) Tanvi** — order is Tanvi $\rightarrow$ Rohan $\rightarrow$ Simmi, so Tanvi is first.
5. **C) East** — the 5 m North and 5 m South cancel out; only 3 m East remains.
6. **C) 162** — each term is $\times 3$: 2, 6, 18, 54, 162.
7. **B) Saturday** — 10 days = 7 + 3; after a full week it's Wednesday again, then +3 more days $\rightarrow$ Saturday.
8. **B) G** — A(1) B(2) C(3) D(4) E(5) F(6) G(7).
9. **D) 42** — 16, 25, 36 are perfect squares (4^2 , 5^2 , 6^2); 42 is not.
10. **B) 12** — each row's two addends rise by 1 and the sum rises by 2: 8, 10, 12.

Section 2 — Mathematical Reasoning

11. **A) 6245** — $3456 + 2789 = 6245$.
12. **A) 4546** — $8002 - 3456 = 4546$.
13. **A) 1404** — $234 \times 6 = 1404$.
14. **B) 117** — $936 \div 8 = 117$.
15. **A) $\frac{4}{6}$** — $\frac{4}{6}$ simplifies to $\frac{2}{3}$.
16. **A) $\frac{3}{4}$** — $\frac{3}{4} = 0.75$, $\frac{2}{3} \approx 0.667$, so $\frac{3}{4}$ is greater.
17. **B) 3250 m** — 3 km = 3000 m; $3000 + 250 = 3250$ m.
18. **B) 7 kg 250 g** — $500 \text{ g} + 750 \text{ g} = 1250 \text{ g} = 1 \text{ kg } 250 \text{ g}$; $4 \text{ kg} + 2 \text{ kg} + 1 \text{ kg } 250 \text{ g} = 7 \text{ kg } 250 \text{ g}$.
19. **C) 4** — every corner of a square is a right angle.
20. **B) 38 cm** — Perimeter = $2 \times (12 + 7) = 2 \times 19 = 38$ cm.

Section 3 — Everyday Mathematics

21. **A) ₹340** — $500 - 125 - 35 = 340$.
22. **B) 270** — $45 \times 6 = 270$.

23. **B) 57 boxes, 0 left** — $456 \div 8 = 57$ exactly.
24. **B) 6 h 30 min** — 8:15 a.m. to 2:15 p.m. is 6 hours; plus 30 more minutes to 2:45 p.m.
25. **B) 1 L 250 mL** — $250 \text{ mL} \times 5 = 1250 \text{ mL} = 1 \text{ L } 250 \text{ mL}$.
26. **B) ₹151** — $(3 \times 45) + (2 \times 8) = 135 + 16 = 151$.
27. **C) 90 L** — $1/4$ of 120 L = 30 L used; $120 - 30 = 90$ L left.
28. **A) 59 kg** — $32 + 27 = 59$.
29. **B) 180** — $15 \times 12 = 180$.
30. **B) 27°C** — $18 + 9 = 27$.

Section 4 — Achievers Section (2 marks each)

31. **B) 165** — Number = divisor $\times$ quotient + remainder = $(7 \times 23) + 4 = 161 + 4 = 165$.
32. **A) 300** — Other number = $1250 - 475 = 775$; difference = $775 - 475 = 300$.
33. **C) ₹375** — Half of 1000 = 500 spent, 500 left; $1/4$ of 500 = 125 spent on books; $500 - 125 = 375$ left.
34. **C) 65** — Bhanu = 12, Aditi = $2 \times 12 = 24$, Chirag = $24 + 5 = 29$; total = $12 + 24 + 29 = 65$.
35. **B) 19** — Each new joined square adds 3 matchsticks to the previous total: for n squares, matchsticks = $3n + 1$. For $n = 6$: $(3 \times 6) + 1 = 19$.

Scoring guide (out of 40): 32+ is Level-2 cutoff range — strong. 24–31 is solid Level-1 performance — review the missed sections. Below 24 — go back to the relevant week(s) in `docs/03-practice-plan.md` before attempting Sample Paper 2.
